EXPERIMENT -10

AIM: Use Wireshark Packet sniffer software and captures TCP, UDP, IP, ARP, ICMP, Telnet, FTP packets with sample output

OBJECTIVE OF THE EXPERIMENT:

Description: Wireshark is a powerful network protocol analyzer that can capture and display data packets traveling through a network. It supports various protocols and can be used to troubleshoot network issues, analyze network performance, and ensure security. Here's an overview of how Wireshark handles different types of packets:

1. **TCP (Transmission Control Protocol)**: Wireshark can capture TCP packets, which are used for reliable, ordered, and error-checked delivery of data between applications. TCP is commonly used for protocols such as HTTP, HTTPS, FTP, and Telnet.
2. **UDP (User Datagram Protocol)**: Wireshark captures UDP packets, which provide a connectionless communication model with a minimal protocol mechanism. UDP is used for time-sensitive transmissions such as DNS queries and streaming media.
3. **IP (Internet Protocol)**: Wireshark captures IP packets, which are used to deliver data across network boundaries. Both IPv4 and IPv6 packets can be analyzed.
4. **ARP (Address Resolution Protocol)**: Wireshark can capture ARP packets, which are used for mapping network addresses (IP) to physical addresses (MAC).
5. **ICMP (Internet Control Message Protocol)**: Wireshark captures ICMP packets, which are used for diagnostic and error-reporting purposes in network operations (e.g., ping).
6. **Telnet**: Wireshark can capture Telnet packets, which provide bidirectional interactive text-oriented communication using a virtual terminal connection over TCP.
7. **FTP (File Transfer Protocol)**: Wireshark captures FTP packets, which are used for the transfer of files between a client and server on a network.

OUTCOME OF THE EXPERIMENT:

With Wireshark, users can inspect individual packets, follow protocol streams, and apply various filters to narrow down the data for more precise analysis. This makes it an invaluable tool for network administrators, cybersecurity professionals, and developers.

PSEUDOCODE:

Wireshark is a powerful tool used for network packet analysis. It allows you to capture and interactively browse the traffic running on a computer network. Below are the steps to capture various types of packets (TCP, UDP, IP, ARP, ICMP, Telnet, FTP) using Wireshark, along with sample outputs.

Steps to Capture Packets Using Wireshark

1. Install Wireshark:
   - Download and install Wireshark from the official website.
2. Start Wireshark:
   - Open Wireshark on your machine.

- Select the network interface you want to capture traffic on (e.g., Ethernet, Wi-Fi).

3. Begin Capture:
   - Click on the interface to start capturing packets.
   - You can stop the capture by clicking on the red square button.
4. Apply Display Filters:
   - Use Wireshark's display filters to focus on specific types of traffic.

RESULTS:

Capturing Specific Packet Types

1. TCP Packets

- **Filter**: tcp
- **Sample Output**:
  plaintext
  Copy code
  No. Time Source Destination Protocol Length Info
  1 0.000000 192.168.1.2 192.168.1.1 TCP 66 55914 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 TSval=12345678 TSecr=0 WS=128

2. UDP Packets

- **Filter**: udp
- **Sample Output**:
  plaintext
  Copy code
  No. Time Source Destination Protocol Length Info
  2 0.001234 192.168.1.2 192.168.1.3 UDP 58 50624 → 53 Len=16 63

3. IP Packets

- **Filter**: ip
- **Sample Output**:
  plaintext
  Copy code
  No. Time Source Destination Protocol Length Info
  3 0.002345 192.168.1.2 192.168.1.4 IP 98 192.168.1.2 → 192.168.1.4 [SYN] Seq=0 Win=64240 Len=0

4. ARP Packets

- **Filter**: arp
- **Sample Output**:
  plaintext
  Copy code
  No. Time Source Destination Protocol Length Info
  4 0.003456 192.168.1.1 192.168.1.2 ARP 42 Who has 192.168.1.2? Tell 192.168.1.1

5. ICMP Packets

- **Filter**: icmp

- **Sample Output**:
  plaintext
  Copy code
  No. Time Source Destination Protocol Length Info
  5 0.004567 192.168.1.2 192.168.1.1 ICMP 74 Echo (ping) request id=0x1234, seq=1/256, ttl=64

6. Telnet Packets

- **Filter**: telnet
- **Sample Output**:
  plaintext
  Copy code
  No. Time Source Destination Protocol Length Info
  6 0.005678 192.168.1.2 192.168.1.5 TELNET 74 Telnet Data

7. FTP Packets

- **Filter**: ftp
- **Sample Output**:
  plaintext
  Copy code
  No. Time Source Destination Protocol Length Info
  7 0.006789 192.168.1.2 192.168.1.6 FTP 78 Request: USER anonymous

Pre Lab Questions:

1. What is Wireshark?
2. How would you setup wireshark to monitor packets passing through an internet router
3. Is it possible to start wireshark from command line on Windows

Sample Viva Voce Questions:

1. Which wireshark filter can be used to check all incoming requests to a HTTP Web server
2. Which wireshark filter can be used to monitor outgoing packets from a specific system on the network.
3. How to capture packets using Wireshark in a switched ethernet network?